Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 5

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : -0.000000
S   : -0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.629656
<D> / site : 0.271609

THIS PDF

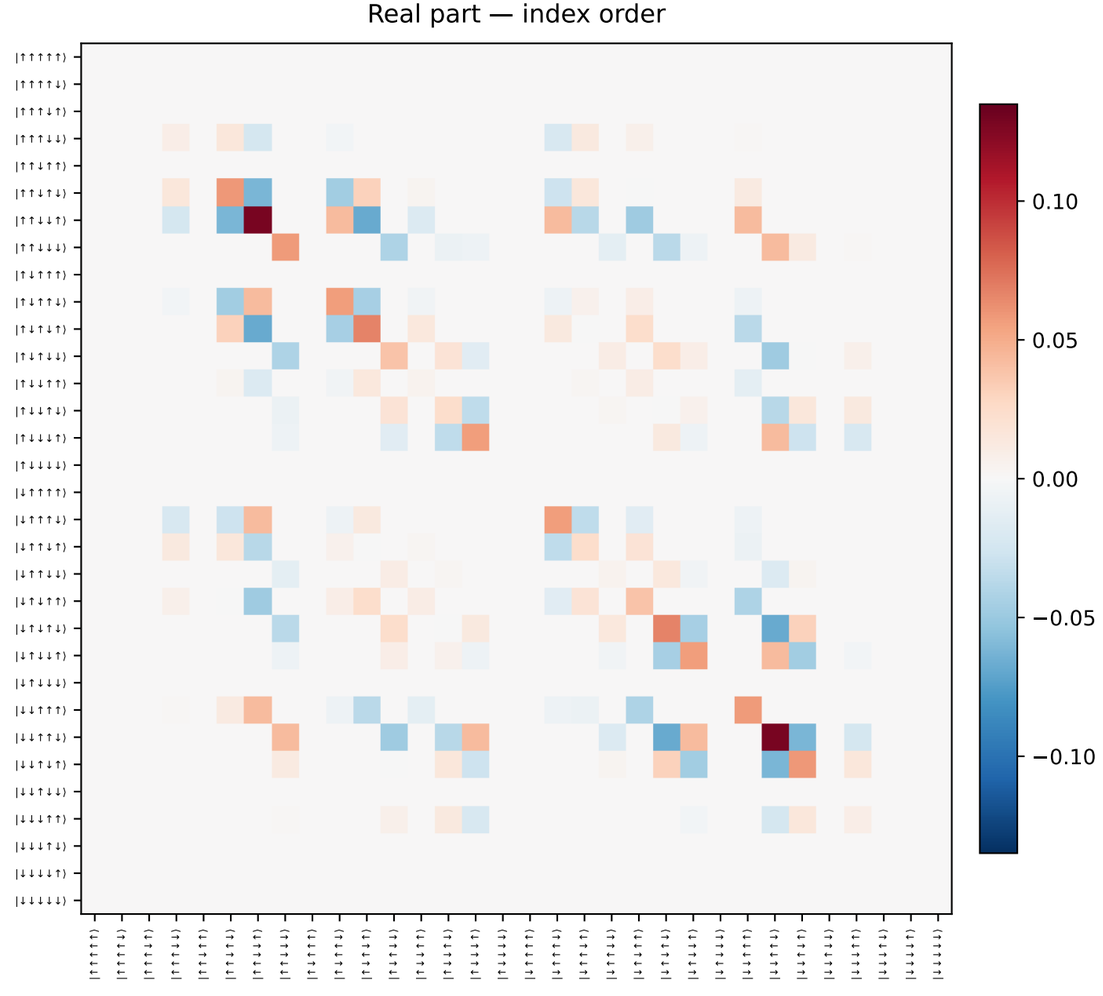
Body order : 5
Clusters   : 6

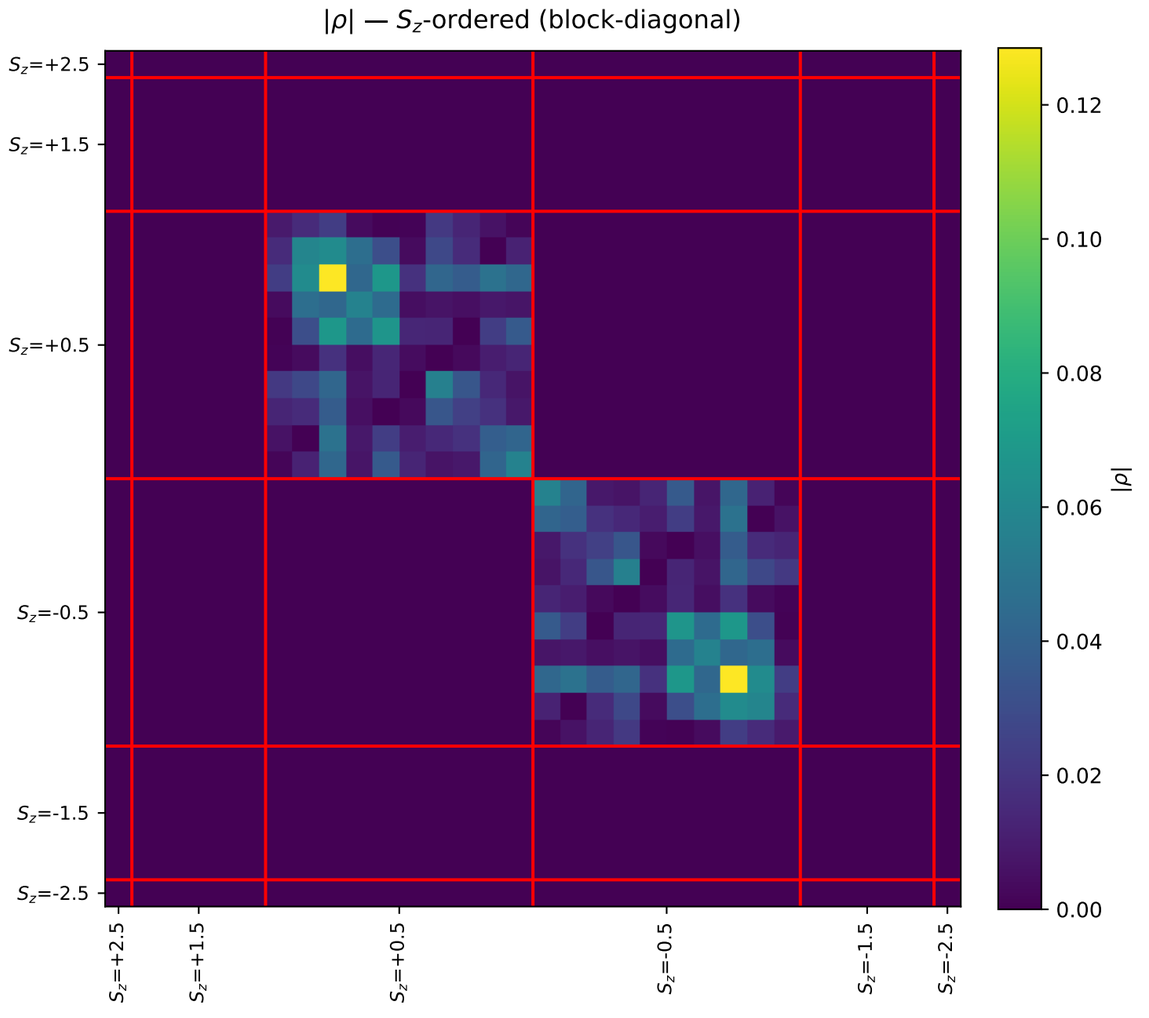
Date: 2026-07-13 00:35

Summary — 5-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3,4	0.200997	1.713944	—	0.000000	0.005144	0.492925	MIXED
0,1,2,3,5	0.283202	1.449504	—	0.000000	0.116381	0.503974	MIXED
0,1,2,4,5	0.296324	1.482678	—	0.000000	0.199760	0.545707	MIXED
0,1,3,4,5	0.315369	1.388419	—	0.000000	0.185343	0.326137	MIXED
0,2,3,4,5	0.196751	1.711856	—	0.000000	0.146413	0.401546	STRONGLY MIXED
1,2,3,4,5	0.259975	1.527688	—	0.000000	0.134886	0.272593	MIXED

5-Body RDM (Sites 0,1,2,3,4)

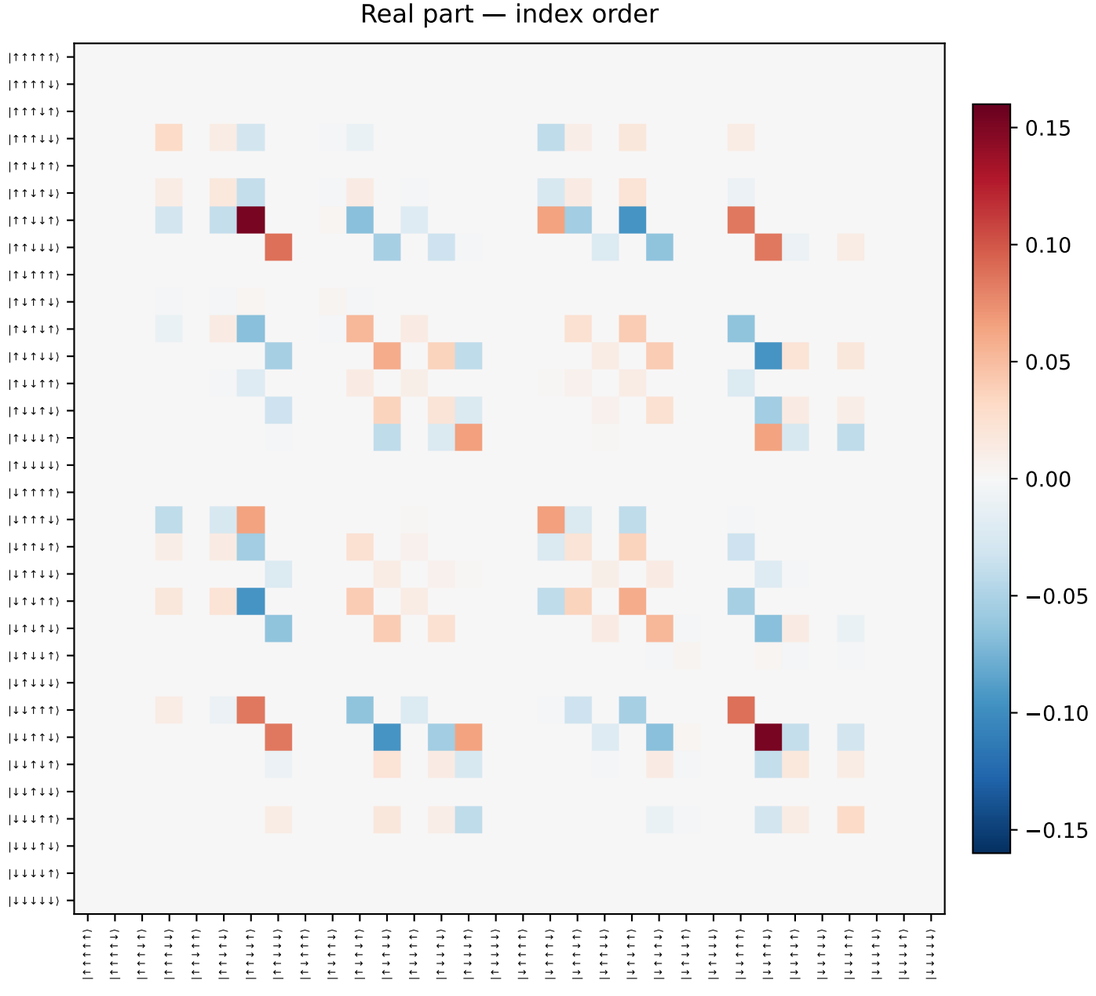


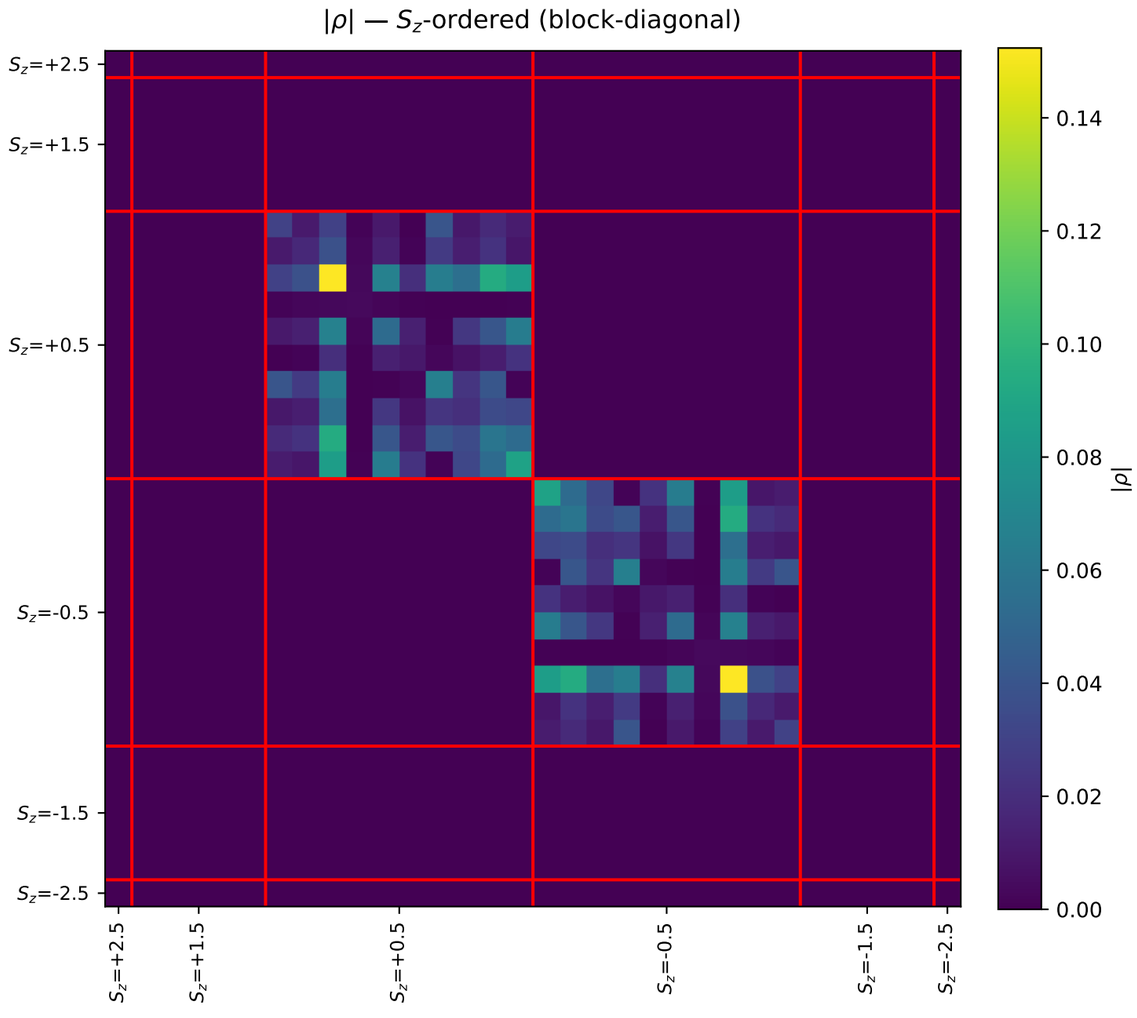


Trace=1.000000 Purity=0.200997 Entropy=1.713944 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.005144 ≤ C_{UB} = 0.492925 verdict: GME

5-Body RDM (Sites 0,1,2,3,5)

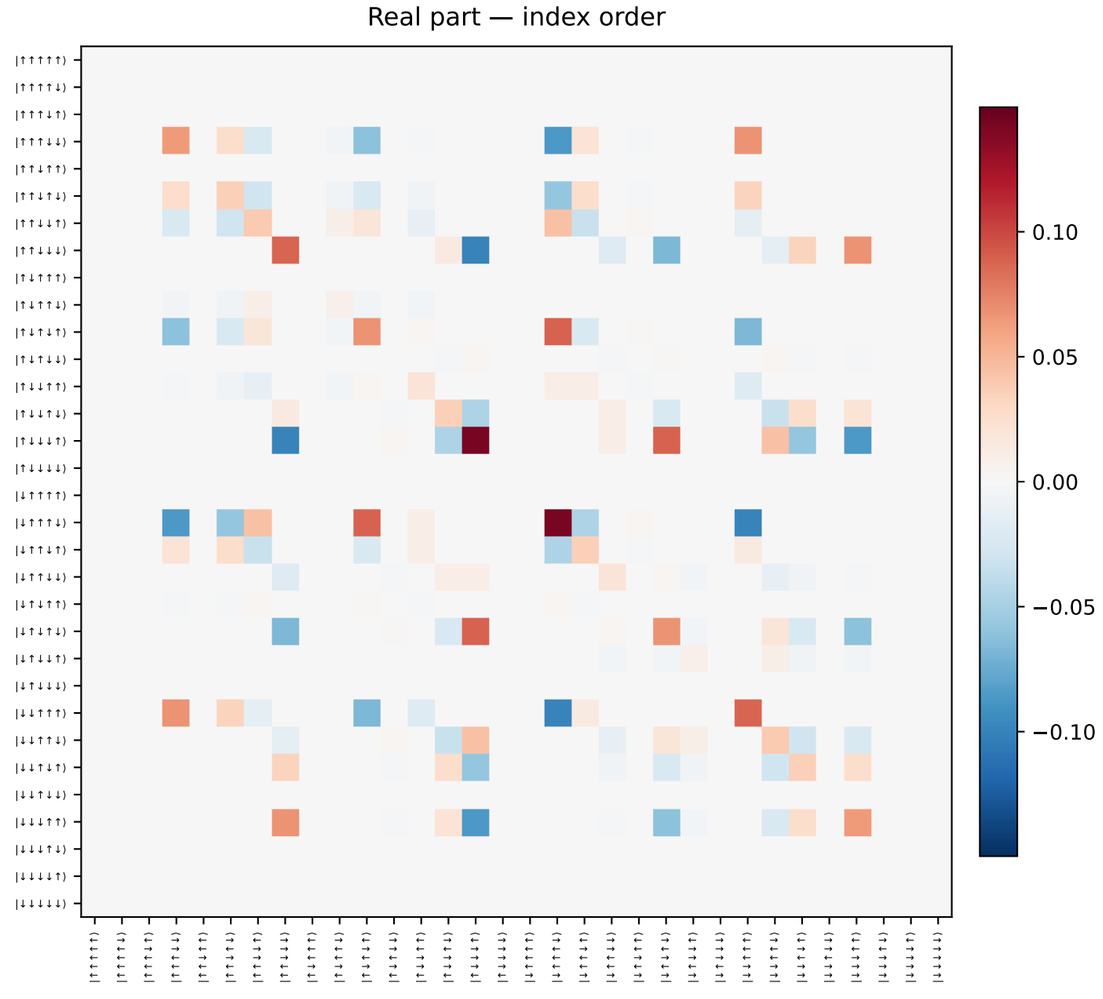


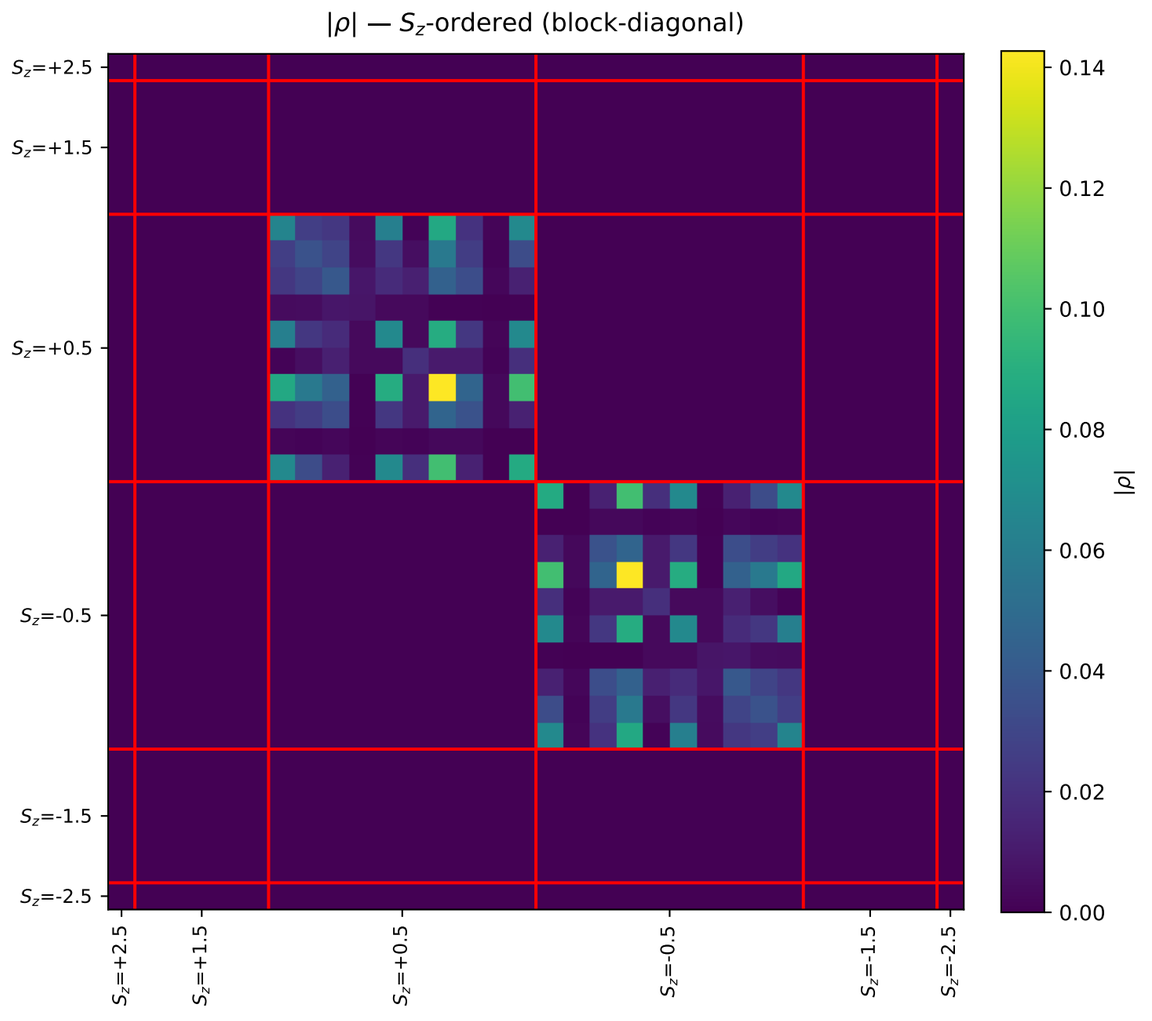


Trace=1.000000 Purity=0.283202 Entropy=1.449504 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.116381 ≤ C_{UB} = 0.503974 verdict: GME

5-Body RDM (Sites 0,1,2,4,5)

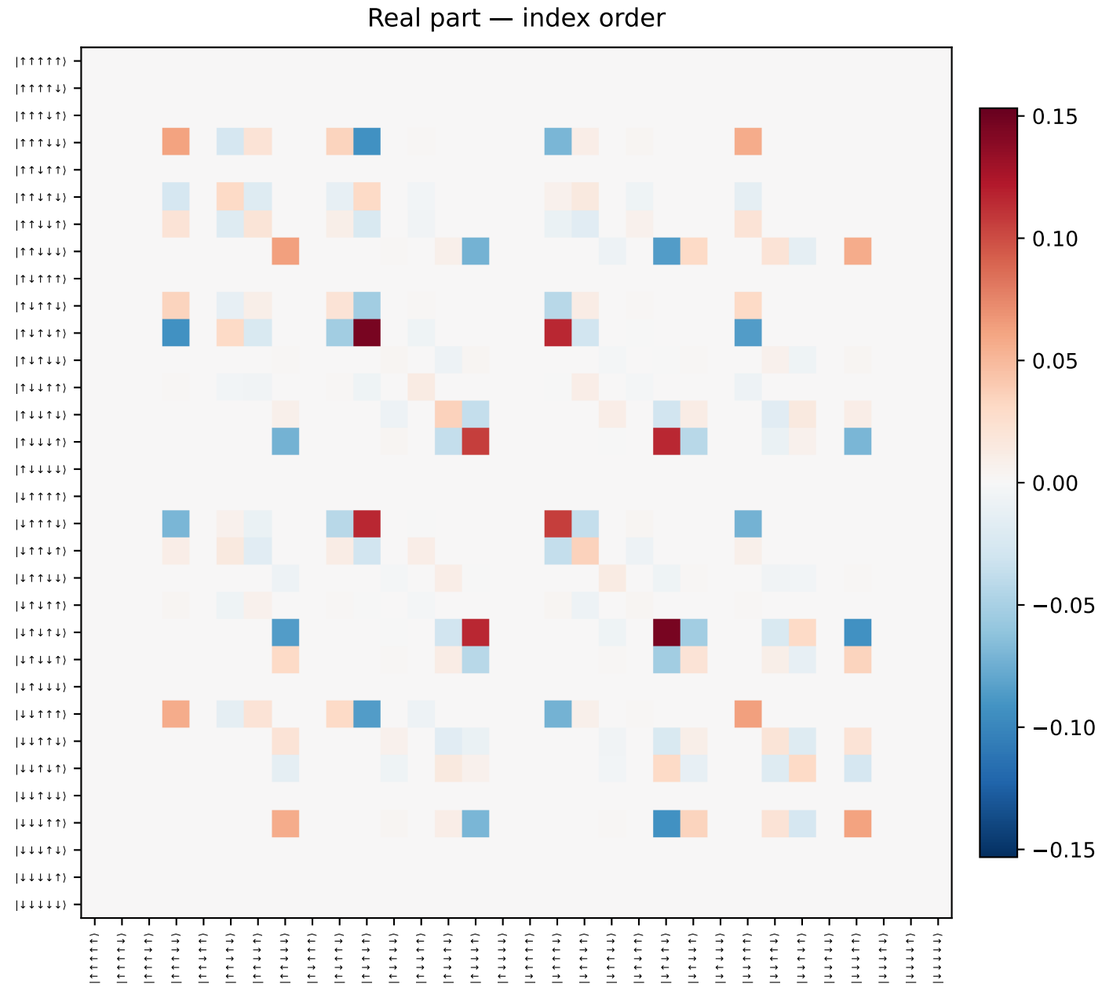


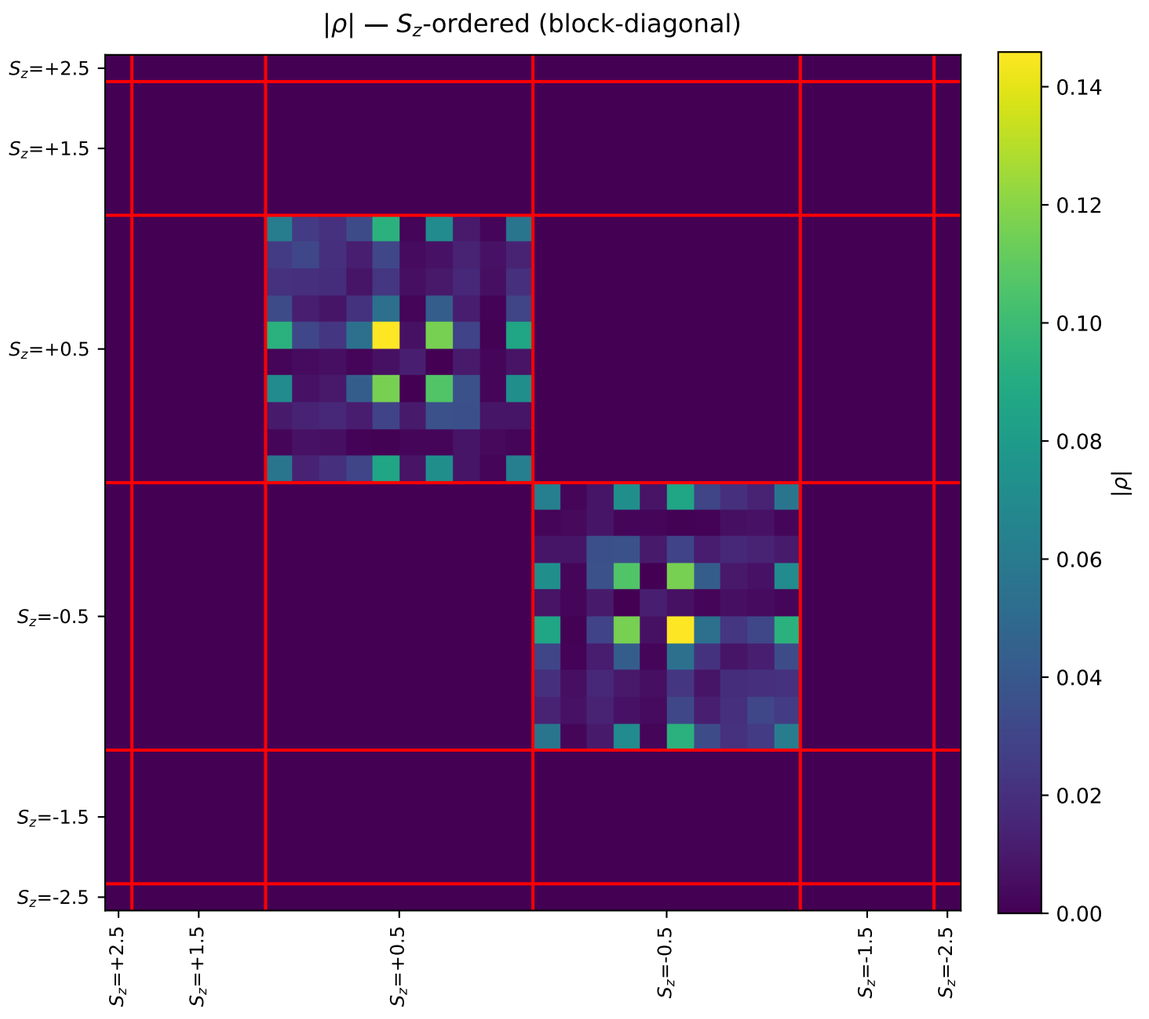


Trace=1.000000 Purity=0.296324 Entropy=1.482678 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.199760 ≤ C_{UB} = 0.545707 verdict: GME

5-Body RDM (Sites 0,1,3,4,5)

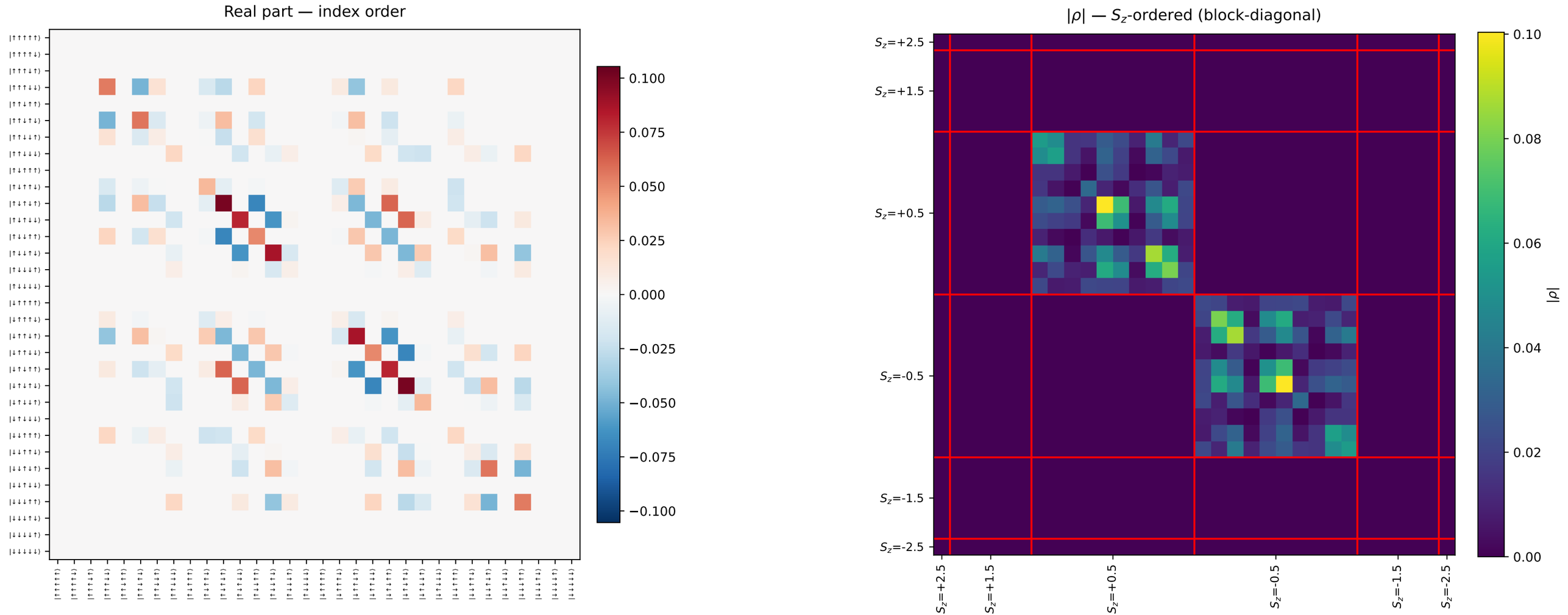




Trace=1.000000 Purity=0.315369 Entropy=1.388419 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.185343 ≤ C_{UB} = 0.326137 verdict: GME

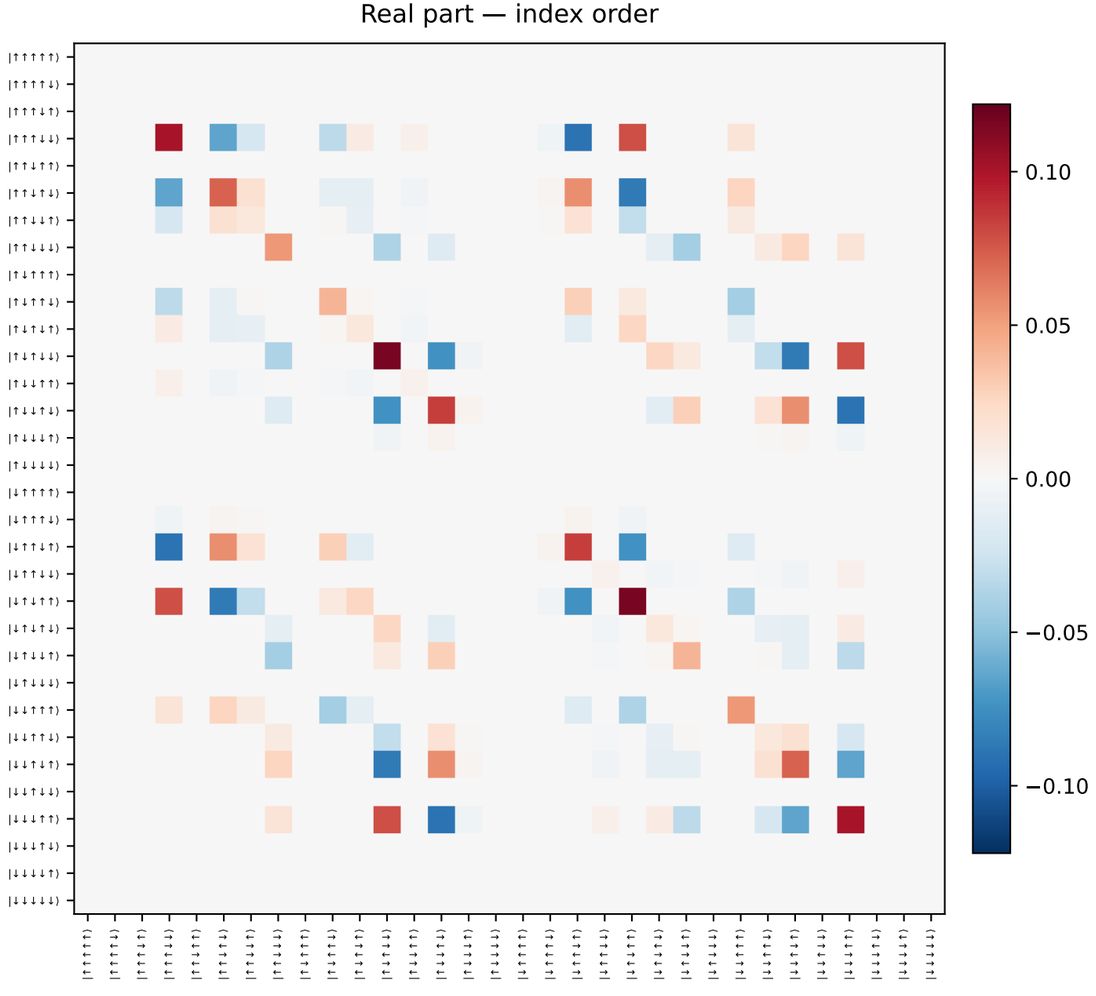
5-Body RDM (Sites 0,2,3,4,5)

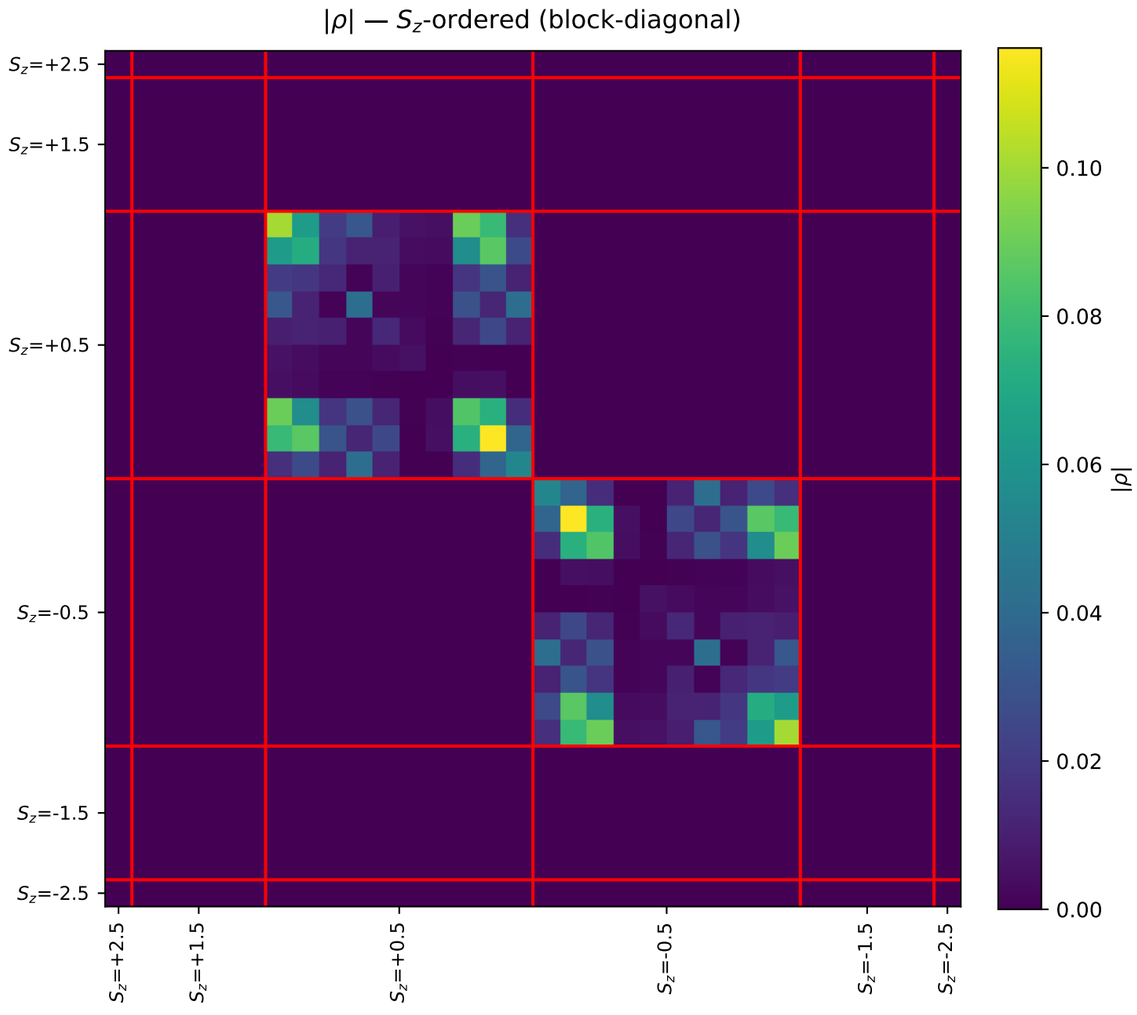


Trace=1.000000 Purity=0.196751 Entropy=1.711856 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.146413 ≤ C_{UB} = 0.401546 verdict: GME

5-Body RDM (Sites 1,2,3,4,5)





Trace=1.000000 Purity=0.259975 Entropy=1.527688 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.134886 ≤ C_{UB} = 0.272593 verdict: GME